MILITARY SPECIFICATION

STEEL, STAINLESS, PRECIPITATION HARDENING [AM 355 (UNS S35500)]

This specification is approved for use by the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the chemical and mechanical properties, metallurgical parameters, and inspection requirements for a precipitation hardening stainless steel commercially known as grade AM 355. Steel shall be produced in the form of forging stock, forgings, mechanical tubing and flash butt welded rings.

1.2 Classification. Grade AM 355 steel shall be of one of the types listed below:

Type I	- Standard Quality
Type II	- Aircraft Quality

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS
FEDERAL

FED-STD-151	- Metals, Test Methods.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-45662	- Calibration Systems Requirements.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to te extent specified herein.

PURCHASE DESCRIPTION
ARMY

M1PD-T-62435	- Trace Elements, Inspection and Test for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

AMS 2241	- Tolerances, Corrosion and Heat Resistant Steel, Iron Alloy, Titanium Alloy Bars and Wire.
AMS 2243	- Tolerances, Corrosion and Heat Resistant Steel Tubing.
AMS 2248	- Chemical Check Analysis Limits, Wrought Corrosive and Heat Resistant Steels and Alloys, Maraging, and other Highly Alloyed Steels, and Iron Alloys.
AMS 2300	- Premium Aircraft - Quality Steel Cleanliness Magnetic Particle Procedure.
AMS 2808	- Identification - Forgings.
AMS 5743	- Bars, Forgings, Forging Stock and Mechanical Tubing.

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AMS 7490	- Rings, Flash Welded, Corrosion and Heat Resistant Austenitic Steels and Austenitic Type Alloys.
AS 478	- Identification and Marking.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

ASTM A604	- Macroetch Testing of Consumable Electrode Remelted Steel Bars and Billets Etchment.
ASTM E3	- Metallographic Specimens, Methods of Preparation.
ASTM E8	- Tensile Testing of Metallic Materials.
ASTM E18	- Rockwell Hardness Test of Metallic Materials.
ASTM E112	- Determining Average Grain Size, Methods for.
ASTM E407	- Microetching Metals and Alloys, Methods for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified, the contractor shall furnish material which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the material to be furnished to the Government. All subsequent materials delivered to the Government shall conform to these samples in all their pertinent physical and performance attributes.

3.2 Materials. Material shall be as specified herein and shall be uniform in quality, condition, cleanliness and soundness. Products specified in highly stressed parts (type II) requiring strength at moderate and high temperatures shall conform to the cleanliness requirements of AMS 2300. They shall be free from foreign materials or internal or external imperfections detrimental to the fabrication or performance of the finished product in accordance with AMS 5743 (see 6.3).

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3.2.1 Chemical composition. Chemical composition of the material shall conform to the percentages by weight for each element specified herein. Specified chemical composition requirements shall apply to standard and aircraft quality steels (see 4.6.2).

TABLE I. Chemical composition.

Elements	Percentage by weight	
	minimum	maximum
Carbon	0.10	0.15
Manganese	0.50	1.25
Silicon	--	0.50
Phosphorus	--	0.025
Sulfur	--	0.025
Chromium	15.00	16.00
Nickel	4.00	5.00
Molybdenum	2.50	3.25
Nitrogen	0.07	0.13
Other elements 1/	--	0.50
Iron	Remainder	

1/ "Other elements" refers to the sum percentage of individual elements present in the chemical composition. The contractor shall analyze for impurities (elements) which may be present in the raw material or occur due to the manufacturing process which affect the product significantly. Test methods and procedures shall conform to the requirements specified in MIPD-T-62435.

3.2.2 Chemical composition variation. Chemical composition variations shall meet the requirements of AMS 2248 (see 4.6.3).

3.2.3 Condition. Unless otherwise specified, the material shall be delivered in the following condition (see 4.6.9):

a. Forging stock shall be suitable for ultrasonic inspection. Rounds shall have smooth surfaces and squares shall have broken corners.

b. Forgings shall be delivered in the heat treated condition. All forgings shall be delivered with a sand or grit blasted surface.

c. Mechanical tubing shall be delivered in the heat treated condition.

d. Rings shall be flash butt welded as specified in AMS 7490 followed solution treated, subzero cooled, equialized and overtempered.

3.3 Tolerances. Acceptable tolerances for specific steel product are as follows (see 4.6.10):

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a. Unless otherwise specified on the purchase order or on the engineering drawings, tolerances on bars shall conform to AMS 2241.

b. Tolerances of forgings and butt welded rings shall be in accordance with applicable engineering drawings.

c. Unless otherwise specified on the purchase order or on the engineering drawings, tolerances on mechanical tubing shall conform to AMS 2243.

3.4 Performance.

3.4.1 Typical mechanical properties. Typical mechanical properties of the material shall be as follows.

3.4.1.1 Tensile properties. Tensile properties of the heat treated material shall be specified in table II (see 4.6.4).

TABLE II. Tensile properties.

	Type I	Type II
Ultimate strength, pound per square inch (psi)	165,000	170,000
Yield strength, at 0.2 percent offset, psi	145,000	155,000
Elongation, in 2 inches of gage length, percent	12	12
Reduction of area, percent	25	25

3.4.2 Hardness. Hardness of the heat treated steel shall be in the range of Hardness Rockwell "C" scale (HRC) 37 to 44 (see 4.6.5).

3.4.3 Metallurgical parameters. Metallurgical properties of the material shall be as specified herein.

3.4.3.1 Microstructure (see 4.6.6).

a. The microstructure of the material shall be essentially tempered martensite.

b. There shall not be more than 15 percent retained austenite when viewed at an appropriate magnification (percentage may vary according to the etching techniques). The percentage, as revealed by X-ray diffraction method shall be correlated with the metallographic method.

c. A small amount of precipitated carbides shall be permitted within the tempered martensite grains. Intergranular oxidation and carbide precipitation shall not be permitted.

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3.4.3.2 Grain size. The average grain size of the material shall be ASTM #4 or finer. Clusters of large germinated grains shall not be acceptable (see 4.6.7).

3.4.3.3 Macrostructure. Presence of inclusions, segregation, stringers, laps and seams shall not be acceptable (see 4.6.8).

3.5 Identification and marking. Identification and marking shall be as specified herein. When marking is performed directly on the material, marking fluid shall have no deleterious effect on the material, and shall be sufficiently stable to withstand normal handling. Marking fluid shall be capable of being removed in a hot alkaline solution (see 4.6.11).

3.5.1 Bars.

a. Each straight bar 0.500 inch and over in diameter, or the least width of flat surface, shall be marked as follows:

   1. Characters shall be marked in a row at intervals of no greater than 3 feet.
   2. Characters shall be of such size as to be clearly legible.
   3. The marking shall include the following data: designated acquisition document and revision letter, heat number, nominal size, and manufacturer's identification.

b. Straight bars less than 0.500 inch in diameter shall be bundled and identified with a metal or plastic tag. The identification tag shall be embossed with the purchase order number, designated acquisition document and revision letter, heat number, nominal size, and manufacturer's identificaton.

c. Coiled bars shall be securely bundled and identified by an attached metal or plastic tag embossed with the purchase order number, designated acquisition document and revision letter, heat number, nominal size, and manufacturer's identification.

3.5.2 Forgings. Forgings shall be identified in accordance with AMS 2808.

3.5.3 Mechanical tubing and flash butt welded rings. Mechanical tubing and flash butt welded rings shall be marked in accordance with AS 478.

3.6 Certification. Unless otherwise specified (see 6.2), the supplier shall furnish 3 copies of forging stock certificaiton with each lot stating the following information.

a. Purchase order number.
b. Title, part number, and date of specification.
c. Quantity.
d. Material specification designation.
e. Heat number.
f. Chemical analysis.
g. Mechanical properties.

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3.7 Workmanship. Material shall be uniform in qualities and condition; clean, sound, and free from foreign materials and from internal and external imperfections detrimental to fabrication or performance of parts. Workmanship shall meet all requirements specified herein and shall assure a product free of defects (see 4.6.13).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection equipment. Unless otherwise specified in the contract, the supplier is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Commercial, modified commercial, or supplier designed inspection equipment or measuring set-ups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2 Materials. To determine conformance to 3.2, inspection, test, and material certification records shall be maintained by the contractor and shall be examined by the Government. Records shall be subject to review by the Government and shall include date, part, or characteristic identification, inspection results, and disposition of lot (accepted or rejected). Corrective action taken on noted defects shall be subject to approval by the Government.

4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).
    1. Examination (see 4.5.2).
    2. Acceptance tests (see 4.5.3).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test condition specified in this specification.

4.4 First article inspection. Unless otherwise specified, the contractor shall supply to the Government the first 2 forgings produced under the contract for inspection to determine conformance to the requirements of this specification. All inspections shall be done by the contractor under Government surveillance.

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4.4.1 Failure. Failure of any first article sample to pass specified examinations or tests shall be cause for refusal to grant first article approval until corrective action by the contractor has been approved by the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all forging stock and forgings of the same heat and condition, from an identifiable production period submitted at one time for acceptance.

4.5.1.2 Sampling for examination. Sampling for quality conformance examination shall be selected in accordance with general inspection levels II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Samples selected in accordance with 4.5.1.2 shall be examined for conformance to the following acceptable quality levels (AQLs) on the basis of percent defective.

Classificaton	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of inspection
Major:	AQL 1.0% Defective	
101	Tolerances, incorrect.	SIE 1/
102	Marking, incorrect or illegible.	Visual
Minor:	AQL 2.5% Defective	
201	Certification, incorrect.	Visual
202	Workmanship, faulty.	Visual
203	Packaging, incorrect.	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Acceptance tests.

4.5.3.1 Sampling. Samples selected in accordance with 4.5.1.2 shall be subjected to acceptance tests specified in table IV.

4.5.3.2 Failure. Any lot that fails to pass any test shall be rejected.

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4.5.3.3 Disposition of rejected material. Material rejected as a result of acceptance test failure may be reworked and resubmitted for inspection.

TABLE IV. Classification of tests.

Description	Requirements	First article	Acceptance tests
Chemical analysis 1/	3.2.1	4.6.2	X
Check analysis 1/	3.2.2	4.6.3	X
Tension test	3.4.1.1	4.6.4	X
Hardness	3.4.2	4.6.5	X
Microstructure	3.4.3.1	4.6.6	X
Grain size	3.4.3.2	4.6.7	X
Macrostructure 1/	3.4.3.3	4.6.8	X

1/ Only these tests shall be performed on forging stock.

4.6 Method of inspection.

4.6.1 Materials. To determine conformance to 3.2, material shall be inspected in accordance with AMS 2300 and AMS 5743.

4.6.2 Chemical analysis. To determine conformance to the requirements of 3.2.1, a sample shall be taken from each lot for each heat of material and shall be analyzed in accordance with MIL-STD-151. Carbon shall be determined by an approved combustion procedure.

4.6.3 Check analysis. To determine conformance to 3.2.2, the sample shall be analyzed in accordance with AMS 2248.

4.6.4 Tension test. To determine conformance to the requirements of 3.4.1.1, the test sample shall be prepared and tested in accordance with ASTM E8.

4.6.5 Hardness. To determine conformance to the requirements of 3.4.2, the sample shall be tested in accordance with ASTM E18.

4.6.6 Microstructure. To determine conformance to the requirements of 3.4.3.1, the metallographic specimens shall be prepared in accordance with ASTM E3 from the heat treated material. Microstructure shall be evaluated by etching the specimen in accordance with ASTM E407 and examining at an appropriate magnification. Material with detrimental microstructure abnormalities shall not be accepted.

4.6.7 Grain size. To determine conformance to 3.4.3.2, specimens shall be tested in accordance with ASTM E112.

4.6.8 Macrostructure. To determine conformance to the requirements of 3.4.3.3, the sample shall be tested in accordance with ASTM A604.

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4.6.9 Condition. To determine conformance to the requirements of 3.2.2, the products shall be verified by visual inspection in accordance with applicable engineering drawings.

4.6.10 Tolerances. To determine conformance to the requirements of 3.3, the bars shall be inspected in accordance with AMS 2241 or in accordance with applicable engineering drawings. Forging stock shall be varified by visual examination. Forgings and butt welded rings shall be inspected in accordance with applicable engineering drawings. Mechanical tubing shall be inspected in accordance with AMS 2243 or in accordance with applicable engineering drawings.

4.6.11 Identification and marking. To determine conformance to the requirements of 3.5, visual examination shall be performed on the identification markings on the products.

4.6.12 Certification. To determine conformance to the requirements of 3.6, visual examination shall be performed on certification.

4.6.13 Workmanship. Workmanship shall be performed at all phases of fabrication and testing to varify conformance to 3.7.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. Two distinct qualities of AM 355 grade stainless steel are described to as standard and aircraft quality in this specification. Standard quality AM 355 shall be used for moderately stressed parts requiring strength and corrosion resistance at moderate temperatures. Aircraft quality steel shall be used for highly stressed parts requiring strength at moderate and high temperatures.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, part number, and date of this specification.
b. First article if specified (see 3.1).
c. Quantity and type of material (see 3.2.3).
d. Certificate in triplicate, if required (see 3.6).
e. Government furnished equipment, to be supplied, if specified (see 4.1.1).
f. Selection of applicable levels of preservation, packaging, and packing (see 5.1).

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6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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